

Table of Contents

1. Executive Summary	3
2. Critical Issues	3
2.1 Broken Email Links	3
2.2 Security: X-Frame-Options Misconfiguration	3
2.3 Contact Form Does Not Submit Data	4
2.4 Missing Legal Pages (GDPR Compliance)	4
2.5 Missing Open Graph Image	4
3. High Priority Issues	5
3.1 Three-Way Pricing Inconsistency	5
3.2 Agent Count and Naming Inconsistency	5
3.3 Services Page Missing from Navigation	5
3.4 Chat AI Describes Wrong Agents	5
3.5 ignoreBuildErrors: true	6
4. Medium Priority Issues	6
4.1 Emoji Usage in Dashboard Components	6
4.2 Dashboard KPI Colour Mismatch	6
4.3 Unused next-intl Dependency	6
4.4 Outdated OG and Twitter Meta Titles	6
4.5 lang="en" Should Be lang="en-IE"	7
4.6 XSS Risk in Chat Widget	7
4.7 FAQ Schema Not Solar-Specific	7
4.8 Dashboard Font Inconsistency	7

4.9 Missing /pricing Page Reference in Chat	7
5. Low Priority Issues	8
5.1 Generic Package Name	8
5.2 Dashboard Clock Flash	8
5.3 Unused Public Assets	8
5.4 AgentCard Dead Code	8
5.5 img Tags Instead of Next.js Image	8
6. Brand Guidelines Compliance	8
7. Page-by-Page Assessment	9
8. Prioritised Recommendations	9
9. Technical Stack Summary	11

1. Executive Summary

This report presents a comprehensive audit of renewably.ie, an AI-as-a-Service platform designed specifically for solar PV installers in Ireland. The website is built on Next.js 16 with React 19, Tailwind CSS v4, and Framer Motion. It comprises 6 public pages (Home, About, Workforce, Services, Blog, Contact), 8 interactive dashboard mockups for the AI workforce, a live AI chat widget powered by z-ai-web-dev-sdk, and a comprehensive CRM system with 30+ API routes.

The overall architecture is solid, the brand identity is strong and consistent, and the interactive dashboard components on the Workforce page represent a significant competitive differentiator. However, the audit identified 5 critical issues, 5 high-priority issues, 9 medium-priority issues, and 5 low-priority items that require attention. The most urgent problems include broken email links across two pages, a security misconfiguration, missing legal pages required for GDPR compliance, a contact form that silently discards all submissions, and a three-way pricing inconsistency across the site.

Severity	Count	Key Areas Affected
CRITICAL	5	Email links, security, form, legal pages, OG image
HIGH	5	Pricing, agent count, navigation, chat AI, build config
MEDIUM	9	Emojis, colours, dependencies, meta, XSS, etc.
LOW	5	Package name, dead refs, clock flash, favicon

Table 1. Issue severity summary

2. Critical Issues

2.1 Broken Email Links

The email address hello@renewably.com appears on the Home page (line 731) and the Workforce page (line 347), but the correct domain is hello@renewably.ie. This means any visitor clicking the email link on two of the most important pages will be directed to a non-existent domain. This is a direct business impact issue: potential customers trying to reach out will receive bounce messages or dead-end navigation, damaging trust and losing leads. The Services page (line 388) and Contact page correctly use hello@renewably.ie, making the inconsistency even more confusing.

Files affected: src/components/HomePageClient.tsx (line 731), src/components/WorkforcePageClient.tsx (line 347)

2.2 Security: X-Frame-Options Misconfiguration

In `next.config.ts` (line 20), the `X-Frame-Options` header is set to `ALLOWALL`, which completely disables clickjacking protection. This allows any external website to embed `renewably.ie` in an `iframe`, potentially tricking users into performing actions on the site without their knowledge. This is a well-known security vulnerability. The value should be changed to `SAMEORIGIN` for production, which allows only `renewably.ie` itself to embed its own pages in `iframes` while blocking all third-party embeds. Additionally, `Content-Security-Policy` headers should be reviewed and tightened.

File affected: `next.config.ts` (line 20)

2.3 Contact Form Does Not Submit Data

The contact form on the Contact page (`ContactPageClient.tsx`, lines 67-72) simulates a submission with a `setTimeout` of 1500ms and then displays a "Message sent!" success state. However, there is no actual API call, no email sending, and no database storage. Every form submission is silently discarded. Users believe their enquiry has been received, but the data goes nowhere. This is a critical business issue: potential leads filling out the contact form are completely lost. The form collects valuable information (name, email, company, jobs per month, message) that should be persisted to a database, sent to `hello@renewably.ie` via an email API, and ideally fed into the CRM.

File affected: `src/components/ContactPageClient.tsx` (lines 67-72)

2.4 Missing Legal Pages (GDPR Compliance)

The footer links to `/privacy` and `/terms` pages, but neither page exists. Clicking these links results in 404 errors. Under GDPR (General Data Protection Regulation), which applies to all businesses operating in Ireland and the EU, having accessible Privacy Policy and Terms of Service pages is a legal requirement. The absence of these pages is not only a compliance risk but also undermines trust with potential customers who expect to see these links lead somewhere. Furthermore, the chat widget stores conversation data and the contact form collects personal information, making a Privacy Policy even more essential.

File affected: `src/components/Footer.tsx` (lines 20-21)

2.5 Missing Open Graph Image

The `layout.tsx` file references `/og-image.png` (line 32) for social media preview cards, but this file does not exist in the `public/` directory. When any page from `renewably.ie` is shared on Facebook, Twitter/X, LinkedIn, or WhatsApp, the preview will show a broken image or no image at all. Social media previews are critical for driving traffic and establishing credibility. A properly sized OG image (1200x630 pixels) should be created and placed in the public directory.

File affected: `src/app/layout.tsx` (line 32)

3. High Priority Issues

3.1 Three-Way Pricing Inconsistency

The website presents three completely different pricing models depending on which page the visitor is viewing. The Home page PricingSection states the cost is "1,000 - 1,500/month". The Workforce page hero section says "1,000-1,500/mo". But the Services page breaks down individual agent pricing at 30-60/month each, totalling approximately 335/month for all agents. These three figures are dramatically different: a visitor who sees 335/month on the Services page will be shocked when the Home page says 1,000+, and vice versa. This inconsistency undermines trust and confuses the buying decision. One canonical pricing structure must be established and applied consistently across all pages.

Page	Stated Price	Source
Home Page	EUR 1,000-1,500/month	HomePageClient.tsx PricingSection
Workforce Page	EUR 1,000-1,500/mo	WorkforcePageClient.tsx hero text
Services Page	EUR 335/month total	ServicesPageClient.tsx line 209
Chat API	EUR 999/month	api/chat/route.ts system prompt

Table 2. Pricing inconsistency across pages

3.2 Agent Count and Naming Inconsistency

The Home page displays 9 agents (including a "Marketing Agent"), the Workforce page displays 8 agents (CEO, Operations, Customer Support, Grants, Logistics, Permitting, QA, Reporting), and the chat API system prompt describes 9 agents with completely different names (Lead Generation, CRM and Sales, Grants and Financing, Logistics, Permitting and Compliance, Quality Assurance, Support, Reporting). The chat AI will describe agents that do not exist on the website, creating confusion when visitors ask the chatbot about the workforce. A single canonical list of agents with consistent names and descriptions must be established and used everywhere: pages, chat API, marketing materials, and schemas.

3.3 Services Page Missing from Navigation

The /services page exists and contains detailed information about each AI agent, individual pricing, before/after comparisons, and the target customer profile. However, it is not linked in the main navigation header. The Header.tsx nav links array includes Home, About, Workforce, Blog, and Contact, but omits Services entirely. Visitors can only discover the Services page through contextual links on other pages. This is a significant navigation gap that reduces the discoverability of one of the most commercially important pages on the site.

File affected: src/components/Header.tsx (line 9)

3.4 Chat AI Describes Wrong Agents

The chat API system prompt (`api/chat/route.ts`) describes 9 agents using names that do not match the website. The prompt lists "Lead Generation Agent", "CRM and Sales Agent", etc., while the Workforce page shows agents named "CEO", "Operations", "Customer Support", and so on. When a visitor asks the chatbot "What agents do you have?", it will describe agents that do not exist on the website. This creates a confusing experience where the AI contradicts the site content. The system prompt must be updated to use the exact same agent names, descriptions, and count as the Workforce page.

3.5 ignoreBuildErrors: true

The `next.config.ts` has `ignoreBuildErrors` set to `true`, which silently suppresses all TypeScript errors during production builds. While this prevents build failures, it also means genuine type errors, missing imports, and broken type contracts can slip into production unnoticed. This is particularly dangerous for a codebase of this size with 21 custom components and 30+ API routes. The flag should be set to `false`, and all existing TypeScript errors should be resolved.

4. Medium Priority Issues

4.1 Emoji Usage in Dashboard Components

All 8 dashboard components (`MiniDesktop`, `OperationsDashboard`, `GrantsDashboard`, `LogisticsDashboard`, `PermittingDashboard`, `QADashboard`, `ReportingDashboard`, `SupportDashboard`) extensively use emoji characters for agent icons, status labels, taskbar icons, and panel headers. The brand guidelines explicitly state "No surrogate pair emojis" because they render inconsistently across platforms. On Windows, many of these emojis will appear as empty boxes or incorrect symbols. All emoji usage should be replaced with white SVG icons or simple text labels, as specified in the brand guidelines ("White taskbar icons in dashboards").

4.2 Dashboard KPI Colour Mismatch

All dashboard components use the colour `#F2CC2E` for KPI value numbers, progress bars, and status indicators. However, the brand primary yellow is `#F3D840`. While the difference is subtle (the dashboard colour is slightly more orange/amber), it creates a perceptible inconsistency when comparing dashboard mockups to the rest of the website. All instances of `#F2CC2E` in dashboard components should be updated to `#F3D840` for brand consistency.

4.3 Unused next-intl Dependency

The `next-intl` package is listed in `package.json` but is never imported or used anywhere in the codebase. There is no internationalization routing, no locale detection, and no translated content. This is a dead dependency that adds unnecessary weight to `node_modules` and may confuse future developers into thinking `i18n` is implemented when it is not. It should be removed.

4.4 Outdated OG and Twitter Meta Titles

The Open Graph and Twitter card titles in layout.tsx (lines 56, 69) still use the generic phrase "AI as a Service for Sales, Marketing and Automation". The site has since been repositioned as an "AI Workforce for Solar PV Installers in Ireland" platform. The meta titles should be updated to match the current positioning so that social media previews accurately reflect the site content.

4.5 lang="en" Should Be lang="en-IE"

The HTML lang attribute in layout.tsx (line 154) is set to "en" but should be "en-IE" to reflect the Irish locale. The site uses Irish English spelling ("optimisation", "behaviour"), Irish-specific terminology (SEAI, ESB, microgeneration), Irish place names, and en-IE locale for date formatting in the blog and chat widget. Setting the correct lang attribute improves accessibility for screen readers and ensures proper hyphenation and pronunciation for Irish English.

4.6 XSS Risk in Chat Widget

The ChatWidget.tsx uses dangerouslySetInnerHTML to render AI responses (line 324) with basic markdown formatting. The formatInlineStyles function replaces markdown patterns with raw HTML, but there is no sanitisation of the AI response content before rendering. If the AI response were to contain a crafted HTML string (e.g., a script tag or an img tag with a tracking pixel), it would be rendered as actual HTML. While the risk is low since responses come from a controlled AI prompt, a sanitisation library such as DOMPurify should be added as a safety measure.

4.7 FAQ Schema Not Solar-Specific

The FAQ schema markup in the Home page (page.tsx, lines 42-94) contains generic questions about AI agencies and AI-as-a-Service, such as "What makes Renewably different from other AI agencies?". These questions do not reflect the site actual positioning as a solar industry-specific platform. The FAQ questions and answers should be rewritten to address solar PV installer concerns, such as SEAI grant handling, ESB grid applications, and ROI calculations for solar businesses.

4.8 Dashboard Font Inconsistency

The dashboard components use "Inter" as their font family, while the main website uses Poppins. While this may be an intentional design choice to give dashboards a distinct "app" feel, it means that the text within the dashboard mockups does not match the surrounding website typography. If this is intentional, it is acceptable, but it should be documented as a deliberate design decision. If consistency is preferred, the dashboard font should be changed to Poppins.

4.9 Missing /pricing Page Reference in Chat

The ChatWidget.tsx PAGE_CONTEXT_MAP (line 49) includes a mapping for "/pricing" but no /pricing page exists on the site. This dead context entry has no effect on functionality but should be removed to keep the codebase clean and prevent future confusion.

5. Low Priority Issues

5.1 Generic Package Name

The package.json name field is still "nextjs_tailwind_shadcn_ts", which is the default from the project scaffolding template. It should be renamed to "renewably" for proper identification.

5.2 Dashboard Clock Flash

The OperationsDashboard initialises its clock state to "12:00:00" (midnight). On first render, before the useEffect fires to set the actual time, the dashboard briefly displays this incorrect time. The initial state should be set to the current time to prevent this visual glitch.

5.3 Unused Public Assets

The public/ directory contains several image files that are not referenced anywhere in the codebase: bot-avatar.png, chat-robot.png, robot-3-nobg.png, and various illustration files. These should be removed to keep the project clean and reduce the static asset footprint.

5.4 AgentCard Dead Code

The AgentCard component in WorkforcePageClient.tsx has a fallback else branch (lines 167-181) that references agent.image with a non-null assertion. However, the agents array has no image property, and all 8 agents are handled by specific num checks, so this branch can never be reached. The dead code should be removed or replaced with proper type safety.

5.5 img Tags Instead of Next.js Image

Several components use plain HTML img tags instead of the Next.js Image component, which provides automatic optimisation, lazy loading, and responsive sizing. Affected files include Header.tsx, ChatWidget.tsx (3 instances), and Footer.tsx. While this does not break functionality, it misses out on performance optimisations that Next.js provides.

6. Brand Guidelines Compliance

The following table summarises compliance with the documented brand guidelines for renewably.ie. Each guideline is assessed as PASS (fully compliant), PARTIAL (mostly compliant with some deviations), or FAIL (significant non-compliance).

Guideline	Status	Notes
Primary Yellow #F3D840	PARTIAL	Dashboards use #F2CC2E instead of #F3D840

Dark #0A0A0A / #1A1A1A	PASS	Used consistently across all pages
British/Irish English	PASS	"optimisation", "behaviour", Irish terminology used correctly
No surrogate pair emojis	FAIL	All 8 dashboard components use emoji characters extensively
Inline styles for HMR	PASS	Dashboards use inline styles; pages mix inline + Tailwind
White taskbar icons	FAIL	Emoji used instead of white SVG icons in all dashboards
Pricing EUR 999/month	PARTIAL	Three different prices shown across Home, Services, Workforce
9 AI agents (canonical)	PARTIAL	8 on Workforce, 9 on Home, different names in chat API

Table 3. Brand guidelines compliance summary

7. Page-by-Page Assessment

Page	Rating	Key Strengths	Key Issues
Home	B+	Strong narrative flow, excellent animations, good CTA structure	Wrong email, generic FAQ schema, 9 agents vs 8 on workforce page
About	A-	Strong founder narrative, authentic Irish voice, good values section	No significant issues found
Workforce	A	Outstanding interactive dashboards, Irish names and places	Wrong email, agent count mismatch with Home page
Services	B	Good pricing breakdown, before/after examples, clear targeting	Pricing contradicts Home page, agent list differs from Workforce
Blog	A-	5 full-length articles, technically accurate, Irish English	Could benefit from more posts for SEO
Contact	B-	Polished UI, good form fields, clear expectations	Form does not actually submit any data

Table 4. Page quality assessment

8. Prioritised Recommendations

Based on the findings of this audit, the following actions are recommended in order of priority. Critical and high-priority items should be addressed before the next production deployment. Medium-priority items should be scheduled for the next sprint cycle. Low-priority items can be addressed during regular maintenance windows.

#	Priority	Action	Effort
1	CRITICAL	Fix email from .com to .ie on Home and Workforce pages	5 min
2	CRITICAL	Change X-Frame-Options to SAMEORIGIN in next.config.ts	2 min
3	CRITICAL	Implement actual contact form submission (API + email)	4-8 hours
4	CRITICAL	Create /privacy and /terms pages for GDPR compliance	2-4 hours
5	CRITICAL	Create /og-image.png (1200x630) for social previews	1-2 hours
6	HIGH	Reconcile pricing across Home, Workforce, Services, and Chat	2-3 hours
7	HIGH	Align agent list across all pages and chat API	2-3 hours
8	HIGH	Add /services to main navigation header	5 min
9	HIGH	Update chat API system prompt with correct agent names	30 min
10	HIGH	Set ignoreBuildErrors to false and fix TypeScript errors	2-4 hours
11	MEDIUM	Replace all dashboard emojis with white SVG icons	4-6 hours
12	MEDIUM	Change dashboard KPI colour from #F2CC2E to #F3D840	30 min
13	MEDIUM	Update OG/Twitter titles to solar-specific positioning	15 min
14	MEDIUM	Change lang="en" to lang="en-IE"	2 min
15	MEDIUM	Rewrite FAQ schema with solar-specific questions	1-2 hours
16	MEDIUM	Add XSS sanitisation for chat AI responses	30 min
17	LOW	Remove unused next-intl dependency	5 min
18	LOW	Clean up unused public/ assets	10 min
19	LOW	Remove dead AgentCard fallback code	10 min

Table 5. Prioritised action items with estimated effort

9. Technical Stack Summary

The following table documents the current technology stack and key configuration for the renewably.ie project. This information is useful for onboarding new developers and for reference during future audits or migrations.

Technology	Version	Purpose
Next.js	^16.1.1	Framework (App Router)
React	^19.0.0	UI library
Tailwind CSS	^4	Utility-first CSS
Framer Motion	^12.23.2	Animations
z-ai-web-dev-sdk	^0.0.17	AI chat completions
Prisma	^6.11.1	ORM (CRM database)
Zustand	^5.0.6	State management
Next Auth	^4.24.11	Authentication (CRM)
Recharts	^2.15.4	Charts
Lucide React	^0.525.0	Icon library
Bun	Runtime	JavaScript runtime and package manager
Poppins	Google Fonts	Primary typeface
ReportLab	Python	PDF generation (this audit)

Table 6. Technology stack overview